Supplementary Material
"Solar Geoengineering Governance: A Dynamic Framework of Farsighted Coalition Formation"

Daniel Heyen *Jere Lehtomaa*

A A framework of geoengineering coalition formation

This section presents the main elements of a general coalition formation framework, making extensive use of Ray and Vohra (2015), Ray (2007) and Hyndman and Ray (2007).[5] We will illustrate the components with a simple example.

A.1 Players and states

Consider a set N of countries with cardinality n, an infinite set $t = 0, 1, \ldots$ of periods, and a compact set X of states. A coalition is a non-empty subset of N. A state captures (at the very least, see extensions below) the *coalition structure*, i.e. how countries are arranged into coalitions. Such a coalition structure is described by a *partition* π of the set of all countries $\{1, \ldots, n\}$. We assume all coalitions to be disjoint, such that any country can only belong to a single coalition at a time. [6] For example, in the case of three countries, $N = \{1, 2, 3\}$, the following coalition structures are possible:

$$x^0 = \{\{1\}, \{2\}, \{3\}\}, x^1 = \{\{23\}, \{1\}\},$$
$$x^2 = \{\{13\}, \{2\}\}, x^3 = \{\{12\}, \{3\}\}, x^{GC} = \{\{123\}\},$$

where, for instance, $\{\{12\}, \{3\}\}$ means that country 1 and country 2 form a coalition and country 3 remains outside as a singleton.[7] State x^{GC} denotes the *grand coalition* comprising all countries, and state x^0 consists of only singleton coalitions.

Countries differ in their political **power** $\gamma_i > 0$, as used in Ricke et al. (2013) and Heitzig and Kornek (2018). Political power might correspond to (a combination of) economic power (measured by GDP), military power, or international reputation. We normalize the terms γ_i with $\sum_{i \in N} \gamma_i = 1$, so that each individual term corresponds to the country's share of total world power.

Power plays a dual role in our framework. First, we assume that a coalition determines its level of geoengineering deployment based on its members' relative power. A more powerful country thus has more say on how a coalition sets the global thermostat. Second, we consider a governance scheme with a minimum power threshold $\underline{\gamma}$ required to implement geoengineering. A coalition C can only deploy geoengineering if its total power exceeds the power threshold, $\sum_{i \in C} \gamma_i \geq \underline{\gamma}$. This power threshold is the first (exogenously defined) way to capture geoengineering governance rules. Ricke et al. (2013), for instance, set $\underline{\gamma} = 0.5$. The threshold only reflects a *necessary* lower bound on "big enough even to start it" or "big enough to push it through against others". For instance, military opposition or economic sanctions to prevent geoengineering might only work if the coalitions that deploy geoengineering are sufficiently weak. The power threshold framework is also fairly flexible. It spans cases as diverse as $\underline{\gamma} = 1$, where only the grand coalition can deploy geoengineering, and $\underline{\gamma} = 0$, which does not impose any power restrictions at all.

I would have expected power as means for coalition transition here as well.

[5] We adopt the non-cooperative bargaining approach instead of cooperative blocking as the default form of interaction between coalitions. The reason is that we think it fits better with international negotiations.

[6] Heitzig and Kornek (2018) consider a nested structure of coalitions, and in Vasconcelos et al. (2020) the coalitions can overlap.

[7] By definition, singletons are also considered as coalitions.

For each country there is a continuous one-period payoff function $u_i : X \to \mathbb{R}$ and a (common) discount factor $\delta \in (0, 1)$. The instantaneous payoff for country i, $u_i(x)$, is then fully described by the going state x. This rests on the assumptions on the equilibrium of the underlying geoengineering deployment game and on whether and how coalitions can transfer utility among their members. We describe these assumptions in detail in Appendix B.

A.2 Institutional rules: Protocol and approval committees

A dynamic coalition formation game requires clear institutional rules to govern which changes in the coalition structure are possible at a given period. Some changes might be infeasible due to existing agreements' irreversibility or if the renegotiation requires the approval of some minimum share of affected players. On the other hand, a game where coalitions can invite new members, dissolve, and merge with other groups over time can be a better description for some settings. To consider a general framework that can handle a range of possible institutional settings, we model the evolution of coalition structures as a sequential game of non-cooperative bargaining. Under this approach, one country in each period can propose a new coalition to form, while other countries can either accept or reject the proposition. We next introduce two complementary concepts that determine the rules of such a game. A protocol specifies an active proposer, a set of potential partners, and the order of responders for each period. The proposer sends the proposition to an approval committee – a collection of countries with the power to validate the transition.

The **protocol** determines how reversible the existing agreements are. Each period t starts with the protocol ρ choosing one country S as the active proposer. For instance, the protocol can be a random process, selecting countries with uniform probabilities or weighted according to their share of global political power. The protocol can also depend on the history of the game in arbitrarily complex ways. Along with the active (singleton) proposer S, the protocol specifies the collection of potential partners $P \subseteq N$ to be all countries that can receive proposals at time t. If the agreements are fully reversible, the set of potential partners is always the entire set of countries N. However, for irreversible treaties, the set only consists of countries that have not formed any previous coalitions during the game so far. Finally, for some subset of all potential partners $Q \subseteq P$, S proposes a new coalition R entering into force.[8] Agreements can also be temporary instead of permanently binding. In such a case, the protocol stops selecting any committed countries as new proposers or potential partners until their going agreements have expired.

Approval committees specify the rules for renegotiation and how much power a particular group has on a new state becoming effective. More formally, for any current state x, let $F_A(x) \subseteq X$ be the set of feasible next states that a set of countries A has the power to approve. Any proposition to move from the current state x to some new state y needs approval from A such that $y \in F_A(x)$. Notably, the set of responders in the approval committee A can potentially be much larger than the set of countries invited to move to a new coalition. Consider the example coalition structures above. If the going state is x^3 and country 3 is the active proposer, it might propose moving the system to x^1 where it forms a new coalition with country 2. Whether country 1 first needs to free country 2 from any previous obligations depends on the game at hand through restrictions on the approval committee. A unanimous acceptance by the approval committee moves the game into the new state. [9] Any rejections mean that the state remains unchanged, and the game moves to the next period with a new proposer selected by the protocol. This game continues indefinitely.

Assumptions on the protocol and approval committees reflect the sphere of international agreements that we are interested in and the bargaining tools that countries have at their disposal in the international arena. First, we place no restrictions on what transitions a country can propose. Any player might then suggest other countries to join forces or existing coalitions to

[8] The proposer can always choose to maintain the status quo. That is, proposing an existing coalition that contains S is always trivially accepted and requires no approval committee.

[9] Extensions are also possible where some non-unanimous vote is enough to pass a proposed transition.

disintegrate, even if it is not self a member of those coalitions. Another form of leverage that we allow the countries to have is the possibility to walk out of their current treaties unilaterally without consulting the other members. For more strictly binding agreements, one could assume, for instance, that a country must be on the approval committee for any transition that affects the composition of its current coalition (Hyndman and Ray 2007).

The **history** of the game, H_t, keeps track of previous coalition structures, propositions, and responses. It can also be included in the state representation x together with the coalition structure π. In some instances, the history is merely a bookkeeping item with little impact on the actual bargaining process. But it also enables the implementation of very detailed protocol-approval committee combinations. One potential use case is the introduction of rejector power, where the choice of new proposers depends on the history of recent rejectors (Chatterjee et al. 1993). In an extreme case, the first country to reject a proposition automatically becomes the new proposer, making the rejection of a proposal a much more valuable tool than under a uniformly random protocol.

A.3 Partition functions and cross-coalition externalities

Externalities across coalitions play an essential role in our geoengineering game. For any individual country, the instantaneous payoff depends not only on the composition of its own coalition but also on what other coalitions exist simultaneously within the entire coalition structure. Partition functions enable a flexible representation of these cross-coalition externalities.

As before, let N be the set of all countries and the coalition structure π a partition of N. Define $M(C, \pi)$ to be a mapping that specifies for each coalition $C \in \pi$ a set of payoff vectors in $\mathbb{R}^{|C|}$.[10] That is, if we assume some transferability of utility, the elements in this set reflect the possible ways any coalition can divide its aggregate payoffs between members, given the ambient coalition structure. A partition function game (N, M) then defines the value of each coalition in every ambient coalition structure that it can be an element of.

A consequence of coalitional payoffs depending on the entire coalition structure is that countries' propositions must also depend on what happens elsewhere in the system. In our context, this might mean that a geoengineering coalition conditions its value sharing rule on whether the outside countries form a counter-geoengineering coalition or remain as singletons. Formally, then, any proposition is a tuple $(C, \boldsymbol{u}(\pi))$ where the vector $\boldsymbol{u}(\pi)$ contains a proposed allocation of value across members in a new coalition C for every conceivable partition π that contains C.

The extent to which utility is transferable across countries is context-specific. For international environmental agreements, transfers might well be technically feasible but politically challenging to implement. The additional complexity of partition functions is that any coalition structure π may allow for multiple equilibrium payoff vectors. To simplify the analysis, it is also possible to impose a fixed sharing rule within a coalition as in Bloch (1996). In our game, such fixed rule could be, for instance, proportional to the political power shares, an equal split across members, or even forbidding any within-coalition utility transfers altogether.

A.4 Farsightedness and bargaining friction

The discounting parameter δ serves two roles in our approach. It introduces bargaining frictions if rejected proposals cause delays in payoffs and controls the countries' level of farsightedness concerning possible future transitions. Countries receive an infinite stream of real-time payoffs based on the going state and their static payoff function $u_i(x)$. Their goal is to maximize the expected sum of discounted payoffs:

$$\mathbb{E}_P\left(\sum_{t=0}^{\infty} \delta^t u_i(x_t)\right), \tag{A.1}$$

[10] In a characteristic function game without externalities, the payoffs are independent of the ambient coalition structure.

where P is a stochastic process governing the state transitions (defined in more detail below). An unsuccessful round of bargaining means that any future gains from coalition formation will be delayed and discounted. A value of δ close to 1 implies that countries might be willing to push their payoffs far into the future or even endure temporary losses to better position themselves for future negotiations.

When choosing their propositions and responses, countries consider not just the one-step consequences of their actions but an entire sequence of new propositions, responses, and transitions that their current actions might trigger. A high level of δ implies that countries consider a long chain of possible future events when choosing their strategies. For simplicity, we assume that all countries share the same level of farsightedness.

A.5 Strategy profiles

A strategy profile defines the country-specific decision rules. We assume all strategies to be stationary and Markovian so that all proposals and responses only depend on the going state with no dependence on history or time. All countries have complete information regarding the details of the game and the strategies of other players.

Let $p_i(x, y, A)$ be the probability that country i proposes moving to state y to an approval committee A when chosen as the active proposer under a going state x. Similarly, let $r_i(x, y, A)$ be the probability that country i, if a member of the approval committee, accepts the proposed transition. The strategy profile is a collection of these proposer and responder distributions across all countries, $\sigma = \{p_i, r_i\}_{i=1}^n$. We can then condense all stochastic components of the model into a single process P_ρ^σ that governs the evolution of our geoengineering game. In other words, the probability of any state transition depends on the protocol picking a particular country k to choose a proposition from p_k and all responding countries $j \in A$ replying according to r_j. The value function of any state x for country i under strategy profiles σ and protocol ρ is then:

$$V_i(x) = (1 - \delta)u_i(x) + \delta \sum_{y \in X} P_\rho^\sigma(x, y)V_i(y). \tag{A.2}$$

An equilibrium strategy profile is a collection of strategies such that no country wants to change their own strategy given the strategies of others. The existence of an equilibrium is guaranteed under the conditions of continuous payoff functions $u_i(x)$ and compact state space X (Harris 1985; Ray 2007), but in general the equilibria are not unique [11]. The following two consistency conditions must apply for all players for a strategy profile to constitute an equilibrium (Hyndman and Ray 2007).

Condition 1. Proposer consistency.
$p_i(x, y, A) > 0$ only if $y \in F_A(x)$ and

$$(y, A) \in \underset{(k,Q):k \in F_Q(x)}{\arg\max} \{\Psi(x, k, Q)V_i(k) + (1 - \Psi(x, k, Q))V_i(x)\}$$

Condition 2. Responder consistency.

$$r_i(x, y, A) \begin{cases} = 1, & \text{if } V_i(y) > V_i(x) \\ = 0, & \text{if } V_i(y) < V_i(x) \\ \in [0, 1] & \text{otherwise.} \end{cases}$$

Above, $\Psi(x, k, Q)$ denotes the probability that the approval committee Q approves the proposed transition from state x to k. In words, the first condition states that for a country to propose a transition, there must be no other possible proposals to make that would have a

[11] A stationary Markov equilibrium exists for finite X (Hyndman and Ray 2007, Supplementary notes).

higher expected value of being accepted. The second condition requires that countries never accept proposals that would take them to a state with a lower expected long-term value.

B Payoffs and geoengineering deployment equilibria

The equilibrium of the underlying geoengineering deployment game depends on assumptions on the preferences of countries and the global effects and costs of geoengineering. This section outlines these assumptions.

B.1 Period payoffs are avoided temperature damages

Each country has a baseline temperature T_i^{Base}. Climate change is assumed to cause a uniform temperature increase by $\Delta > 0$

$$T_i^{\text{CC}} = T_i^{\text{Base}} + \Delta. \tag{B.1}$$

Geoengineering reduces global temperatures, and we assume for clarity of the exposition that it reduces temperatures uniformly,

$$T_i = T_i^{\text{CC}} - G, \tag{B.2}$$

where we measure the geoengineering level G in terms of the uniform temperature effect. Following Burke et al. (2015), we assume there is a uniform ideal temperature, T^{ideal}, with climate damages increasing quadratically in the deviation from that ideal temperature,

$$u_i(T_i) = -d_i(T_i - T^{\text{ideal}})^2 + K_i, \tag{B.3}$$

where K_i is a constant. We can rewrite (B.3) as a function of the global geoengineering level,

$$u_i(G) = d_i \cdot G \cdot (2\alpha_i - G), \tag{B.4}$$

where $\alpha_i = T_i^{\text{Base}} + \Delta - T^{\text{ideal}}$ is the ideal SG level of country i and $K_i = d_i(T_i^{\text{Base}} + \Delta - T^{\text{ideal}})^2$ are country i's climate damages at the local temperature caused by climate change. This choice of K_i ensures that we can interpret u_i as country i's benefit from SG. It is expression (B.4) that we use in the main text to calculate period payoffs for countries. In the main text we set $d_i = 1$ and $T^{\text{ideal}} = 13°\text{C}$. The constant term K_i could also capture any within-coalition transfer schemes, which we ignore in this paper.

B.2 Global geoengineering deployment

There are n countries in total. Apart from decisions on coalition formation, which is our main focus, countries choose geoengineering levels g_i. We assume that there are no direct costs of geoengineering. If only solar geoengineering is allowed, then $g_i \geq 0$. With counter geoengineering, $g_i \in \mathbb{R}$. Geoengineering efforts by different countries simply add up to the total global geoengineering level:

$$G = \sum_{i=1}^{n} g_i. \tag{B.5}$$

B.3 Best response functions

Here, we determine the best response function of a coalition. Let $C \subseteq N$ be a coalition that exceeds the minimal power threshold, $\sum_{i \in C} \gamma_i \geq \underline{\gamma}$, and let $G_{-C} \geq 0$ be the amount of cooling provided by all other countries. If the coalition falls short of the minimal power threshold, its best response trivially is zero. The coalition's choice variable is $G_C \geq 0$, and the final total geoengineering level will be $G = G_C + G_{-C}$. The coalition's objective is to maximize the power-adjusted sum of utilities of its members,

$$\max_{G_C \geq 0} \sum_{i \in C} \gamma_i \cdot \left(\kappa_i - d_i(G_C + G_{-C} - \alpha_i)^2\right). \tag{B.6}$$

The constant κ_i is irrelevant for the optimization, so (B.6) is equivalent to maximizing:

$$-\sum_{i \in C} \eta_i (G_C + G_{-C} - \alpha_i)^2, \tag{B.7}$$

where $\eta_i = \gamma_i d_i$ combines a country's political power and climatic exposure, measured in the marginal temperature damage. Expression (B.7) is strictly concave in G_C with a maximum at

$$G_C(G_{-C}) = \max(\bar{\alpha}_C - G_{-C}, 0), \tag{B.8}$$

where

$$\bar{\alpha}_C = (\sum_{i \in C} \eta_i)^{-1} \sum_{i \in C} \eta_i \alpha_i \tag{B.9}$$

is the coalition's *weight-adjusted average ideal level.* Note that (B.9) simplifies to the arithmetic mean if all η_i are the same; also, a singleton's weight-adjusted ideal level is the original ideal level.

B.4 Nash equilibrium

Take any coalition structure $\pi = C_1 \cup \ldots \cup C_K$. Assume the coalitions are ordered according to their weight-adjusted ideal level, i.e. $\bar{\alpha}_{C_1} > \ldots > \bar{\alpha}_{C_K}$. For simplicity, we assume that the $\bar{\alpha}_{C_k}$ are pairwise-different. We can ensure pairwise different weight-adjusted average ideal levels by very small changes to either powers γ_i, marginal damages d_i and/or ideal deployment levels α_i.

We have determined the best response functions in (B.8). All best response functions have the same slope. They don't overlap because the $\bar{\alpha}_{C_k}$ are pairwise different. Taken together, this shows that the unique Nash-equilibrium is:

$$G^*_{C_1} = \bar{\alpha}_{C_1}, \quad G^*_{C_k} = 0 \text{ for } k = 2, \ldots, K. \tag{B.10}$$

This is in line with the free-driver narrative: the coalition of countries with the highest ideal geoengineering level, here captured by the weight-adjusted average ideal level, implements that level and imposes it on everyone else. The total geoengineering level of coalition structure π can be written as:

$$G^*(\pi) = \max_k \{\bar{\alpha}_{C_k} | C_k \in \pi\}. \tag{B.11}$$

C Equilibrium strategy profiles

Here we describe the equilibrium strategy profiles in full detail, corresponding to the two examples in Section 3. As noted in the main text, a strategy implies countries having contingency plans for all possible states where they might be chosen as the proposer, or asked to respond to a proposal as a member of an approval committee. We construct the strategy profiles manually from an initial guess according to the pre-specified approval committee structures, and verify their correctness using Condition 1 and Condition 2 in section A.5.

			Proposer W					Proposer T					Proposer C				
			( )	(TC)	(WC)	(WT)	(WTC)	( )	(TC)	(WC)	(WT)	(WTC)	( )	(TC)	(WC)	(WT)	(WTC)
( )	Proposition	NaN	1					1					1				
( )	Acceptance	W	1							0	0	0			0	0	0
( )	Acceptance	T		0		1	1	1						1		1	1
( )	Acceptance	C		0	1		1		0	1		1	1				
(TC)	Proposition	NaN		1				1					1				
(TC)	Acceptance	W		1						0	0	0			0	0	0
(TC)	Acceptance	T	0		1	1	1	1	1						1	1	1
(TC)	Acceptance	C	0		1	1	1			1	1	1	1	1			
(WC)	Proposition	NaN	1							1					1		
(WC)	Acceptance	W	1		1			1	1		1	0		1		1	0
(WC)	Acceptance	T		0		0	1			1				0		0	1
(WC)	Acceptance	C		0		0	1	0	0		0	1	0		1		
(WT)	Proposition	NaN	1								1					1	
(WT)	Acceptance	W	1			1			1	0		0	1	1	0		0
(WT)	Acceptance	T		0	1		1	0			1		0	0	1		1
(WT)	Acceptance	C		0	1		1		0	1		1				1	
(WTC)	Proposition	NaN		1								1					1
(WTC)	Acceptance	W		1			1	1	1		1		1	1	1		
(WTC)	Acceptance	T	0		0	0				0		1	0	0	0		
(WTC)	Acceptance	C	0		0	0		0	0		0					0	1

Figure 3: Strategy profile under the weak governance scenario. Dark orange tiles indicate transitions that the proposer can initiate without consulting others (maintaining status quo or unilaterally leaving a coalition). Grey tiles indicate that a country is not a member of the corresponding approval committee. Light green tiles are transitions for which a country is in the approval committee. Light orange tiles on the left correspond to the going state of the game. Numerical values indicate proposal and acceptance probabilities between 0 and 1.

			Proposer W					Proposer T					Proposer C				
			( )	(TC)	(WC)	(WT)	(WTC)	( )	(TC)	(WC)	(WT)	(WTC)	( )	(TC)	(WC)	(WT)	(WTC)
( )	Proposition	NaN		1					1					1			
( )	Acceptance	W	1							1	1	1			1	1	1
( )	Acceptance	T		1		0	0	1						1		0	0
( )	Acceptance	C		1	0		0		1	0		0	1				
(TC)	Proposition	NaN					1		1					1			
(TC)	Acceptance	W		1						1	1	1			1	1	1
(TC)	Acceptance	T	0		0	0	0	0	1						0	0	0
(TC)	Acceptance	C	0		0	0	0			0	0	0	0	1			
(WC)	Proposition	NaN			1				1				1				
(WC)	Acceptance	W	0		1			0	0		1	1		0		1	1
(WC)	Acceptance	T		1		0	0			1				1		0	0
(WC)	Acceptance	C		1		0	0	1	1		0	0	1		1		
(WT)	Proposition	NaN				1		1						1			
(WT)	Acceptance	W	0			1			0	0		1	0	0	0		1
(WT)	Acceptance	T		1	1		1	1			1		1	1	1		1
(WT)	Acceptance	C		1	1		0		1	1		0				1	
(WTC)	Proposition	NaN					1			1						1	
(WTC)	Acceptance	W		0			1	0	0		0		0	0	0		
(WTC)	Acceptance	T	1		1	0				1		1	1	1	1		
(WTC)	Acceptance	C	1		1	1		1	1		1					1	1

Figure 4: Strategy profile under the power threshold scenario. Dark orange tiles indicate transitions that the proposer can initiate without consulting others (maintaining status quo or unilaterally leaving a coalition). Grey tiles indicate that a country is not a member of the corresponding approval committee. Light green tiles are transitions for which a country is in the approval committee. Light orange tiles on the left correspond to the going state of the game. Numerical values indicate proposal and acceptance probabilities between 0 and 1.

D Additional examples

For the clarity of exposition, the examples presented in Section 3 assumed that the countries differ only in terms of their ideal temperature levels. We also contrasted the weak governance scenario with free-driver behaviour to only one possible governance scheme with a minimum power threshold. In this supplementary section, we provide two additional examples. First, we consider the same minimum power threshold rule as in the main text but now allow countries to differ in their marginal damages d_i. Then, building on the example with differentiated marginal damages, we consider an alternative set of institutional rules where a unanimous approval committee is no longer needed, but a majority approval is enough to verify a state transition. As we will show, both extensions will directly affect the transition dynamics of the model, and in the latter example, also on the predictions of the absorbing state.

Power threshold with country-specific marginal damages Consider first a setting similar to Section 3 with a minimum power threshold for SG deployment. All countries still have the same ideal temperature levels shown in Table 1. But whereas all marginal damages d_i are fixed to unity in the main text, we now consider $d_W = 0.75$, $d_T = 1.25$, and $d_C = 1.0$. Alternatively, we could introduce variation in the damage functions by considering additional terms (such as precipitation effects) or by considering different marginal impacts from excessive and insufficient levels of geoengineering.

The main consequence of the new marginal damages is that all countries now receive their highest possible period payoffs in different states: country W in (WT), T in the grand coalition (WTC), and C in (TC) (see panel A in Figure 5). The exact same strategy profile shown in

Figure 4 still constitutes an equilibrium in this new example. The key thing to note is that country T still wants to break out of the grand coalition, even though it maximizes the period payoffs for T. This means that country C has a high degree of veto power on the system dynamics through being able to break the grand coalition and by T anticipating its strategy.

Consider the game in (WTC). Country T knows that W remains passive, as the warm country tries to maintain any high-deployment state for as long as possible. The bargaining therefore occurs essentially between the two natural partners T and C, who have relatively similar ideal temperature levels, and together have enough political clout to deploy SG.

Since the countries are identical in power and the proposer protocol is a uniform distribution, there is a 1/3 change of C becoming the proposer in (WTC) and breaking the system into (WT), knowing that from there it is in the interest of T to further deviate into all singletons ( ), again with probability 1/3. Anticipating these moves, and acting first, country T can first deviate from (WTC) to (WC), forcing C to perform the additional deviation to ( ). For T, such a sequence of states implies a higher sum of cumulative payoffs due to the more favourable transition. Therefore, T anticipating the actions of C, it also prefers to break out from the state where the SG deployment level would bring it closest to its ideal temperature, simply in anticipation over the strategies of others.

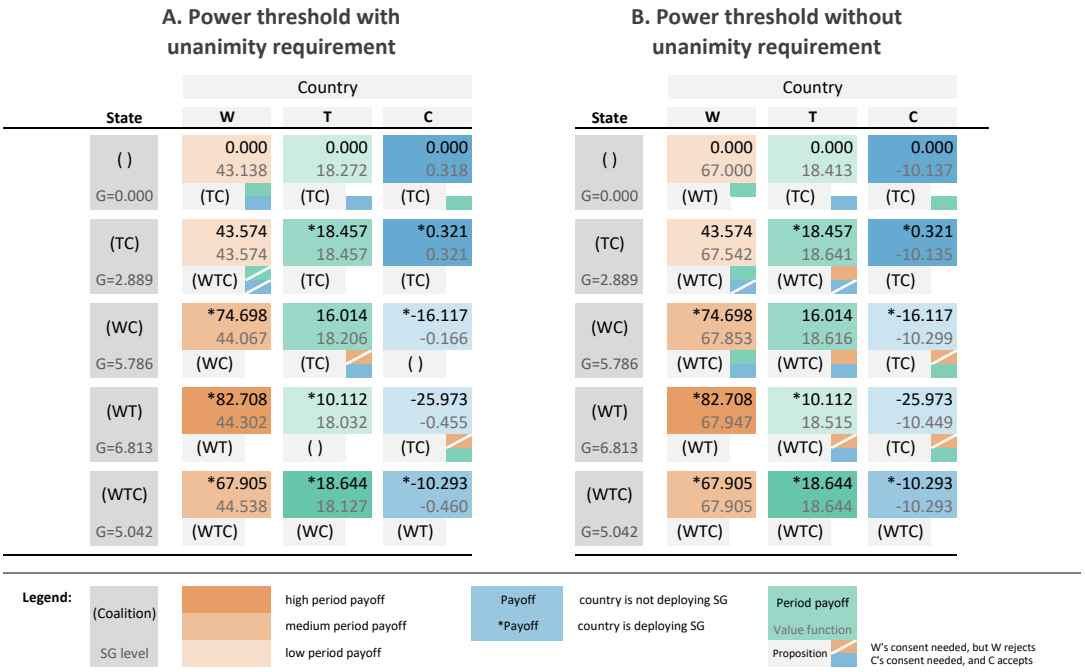

Figure 5: Summary of period payoffs and equilibrium strategy profiles under differentiated marginal damages with the unanimity requirement (panel A) and without it (panel B). Period-payoffs u_i, cf. expression (2), and long-term expected values given strategies V_i, cf. expression (1), are shown in red (country W), green (country T), and blue (country C) boxes. Dark and light fills represent more and less attractive constellations, respectively. An asterisk in front of the payoff indicates that the respective country is deploying SG. Every row is dedicated to a certain going state. The dark grey box on the left indicates the coalition structure and the total global SG level in that state. For every state the light grey boxes show the proposal of the respective country in equilibrium. If that proposition needs approval by others, the countries needed for approval are indicated by small colored boxes to the right of the proposal: solid filled if the respective country approves the proposal in equilibrium, filled with a diagonal line if the respective country rejects the proposal.

Power threshold with country-specific marginal damages and majority approval committees Finally, consider the same setting as above with a minimum power threshold and differentiated marginal damages, but this time assuming that instead of a unanimous approval committee, only 50% of the committee members is required to validate a state transition. More concretely for our game, this implies that all new members must still approve a transition (e.g. W proposing the move from ( ) to (WTC) needs the consent of both T and C), but only half of the existing members (for instance, W can propose, and T can approve the transition from (TC) to (WTC) without consulting country C).

Dropping the unanimity requirement can have a large impact on the transition dynamics. Consider the strategies in panel B in Figure 5, under which the grand coalition (WTC) is an absorbing state. So far, country C always had an incentive to break the grand coalition, expecting T to eventually play along and move the system to the absorbing state (TC). However, without the unanimity requirement, it would be in the interest of T to approve W entering (TC), thus restoring the grand coalition. Since C needs to endure high transitory losses when aiming for (TC) to materialize, this strategy is no longer profitable when the unanimity requirement is dropped.

The resulting transition probabilities are presented in Figure 6, and the detailed strategy profile in Figure 7. Note that all countries now want to avoid the singleton structure ( ). This is intuitive: since it is now easier for countries to trigger state transitions, there is less veto power from dropping out from any particular treaty. All countries anticipate that some SG deploying coalition will eventually emerge, so it is in their interest to join in and have a say on the thermostat setting instead of remaining outside. Note that in this scenario, the geoengineering deployment levels, and hence also the period payoff levels remain unchanged from the panel A in Figure 5 with the unanimity requirement still in place. But the value functions for each state, as described in expression (A.2), change, given the new transition probabilities between states.

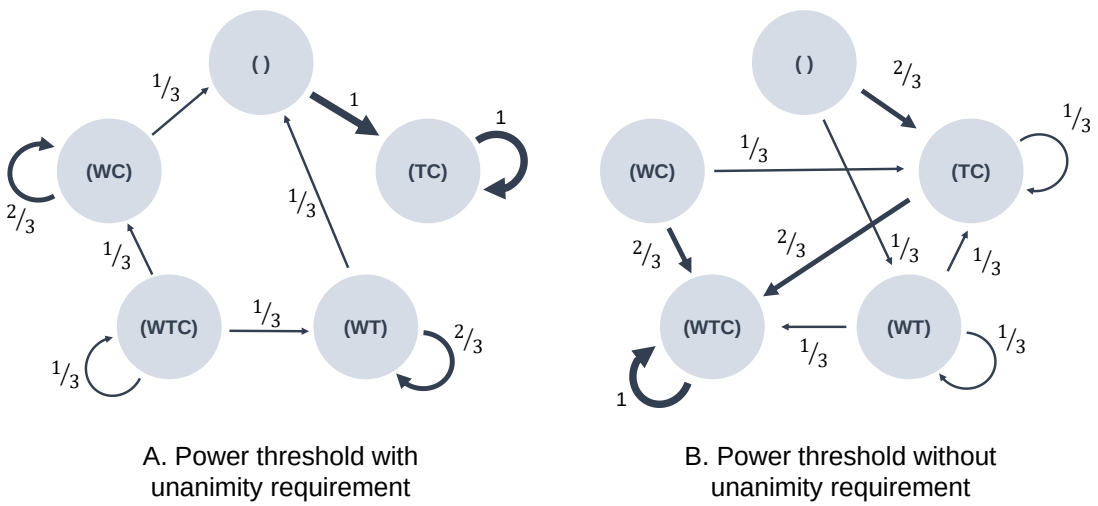

Figure 6: State transition probabilities under power threshold, country-specific marginal damages and unanimity requirement (panel A) and without unanimity requirement (panel B). The values in panel A correspond to the power threshold scenario in Section 3 and strategy profile in Figure 4, repeated here for convenience. The state transition probabilities in panel B follow from the equilibrium strategies shown in Figure 7. Without unanimity requirement the absorbing state is (WTC), as country C no longer has the incentive to deviate, knowing that countries W and T have the power to restore the grand coalition from state (TC).

			Proposer W					Proposer T					Proposer C				
			( )	(TC)	(WC)	(WT)	(WTC)	( )	(TC)	(WC)	(WT)	(WTC)	( )	(TC)	(WC)	(WT)	(WTC)
( )	Proposition	NaN				1			1					1			
( )	Acceptance	W	1							1	1	1			1	1	1
( )	Acceptance	T		1		1	1	1						1		1	1
( )	Acceptance	C		1	0		0		1	0		0	1				
(TC)	Proposition	NaN					1					1		1			
(TC)	Acceptance	W		1						1	1	1			1	1	1
(TC)	Acceptance	T	0		0	0	1	0	1						0	0	1
(TC)	Acceptance	C	0		0	0	0			0	0	0	0	1			
(WC)	Proposition	NaN					1					1		1			
(WC)	Acceptance	W	0		1			0	0		1	1		0		1	1
(WC)	Acceptance	T		1		0	1			1				1		0	1
(WC)	Acceptance	C		1		0	1	1	1		0	1	1		1		
(WT)	Proposition	NaN				1						1		1			
(WT)	Acceptance	W	0			1			0	0		0	0	0	0		0
(WT)	Acceptance	T		1	1		1	0			1		0	1	1		1
(WT)	Acceptance	C		1	1		1		1	1		1				1	
(WTC)	Proposition	NaN					1					1					1
(WTC)	Acceptance	W		0			1	0	0		1		0	0	0		
(WTC)	Acceptance	T	0		0	0				0		1	0	0	0		
(WTC)	Acceptance	C	1		0	0		1	1		0					0	1

Figure 7: Strategy profile without unanimity requirement under a power threshold and differentiated marginal damages. Dark orange tiles indicate transitions that the proposer can initiate without consulting others (maintaining status quo or unilaterally leaving a coalition). Grey tiles indicate that a country is not a member of the corresponding approval committee. Light green tiles are transitions for which a country is in the approval committee. Light orange tiles on the left correspond to the going state of the game. Numerical values indicate proposal and acceptance probabilities between 0 and 1. For the strategy profile under unanimity requirement, see Figure 4.